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Inconsistent and non-complete logics.
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Traditionally, it has been understood that an argument is *valid* iff whenever the premises are true, the conclusion is true as well. Accordingly, it has been understood that an argument is invalid there are *cases* in which the premises are true but conclusion is non-true (or false). From this, in classical logic

1. Every argument is valid or invalid, and
2. No argument is not both valid and invalid.

However, recent literature in philosophical logic has disputed the idea that truth and falsity are mutually exclusive and exhaustive (for example, in [1]). From this it is natural to conjecture that similar failures may occur at the level of argument validity as well.

In this talk, we present and motivate examples of what we call *inconsistent* and *non-complete* logics:

Inconsistent logic: A logic $\mathbf{L}$ is *inconsistent* if there is an argument which is both valid and invalid in $\mathbf{L}$.

Non-complete logic: A logic $\mathbf{L}$ is *non-complete* if there is an argument which is neither valid nor invalid in $\mathbf{L}$.

We do so by modifying the implicit invalidity condition in well-known non-classical logics such as $\mathbf{K3}$, $\mathbf{LP}$, and $\mathbf{FDE}$. Although this may initially sound unusual, the modification we propose is neither bizarre nor *ad hoc*, insofar as our condition is equivalent to the classical invalidity condition. We present two more logics that are both inconsistent and non-complete, by modify the connexive logics $\mathbf{MC}[3]$ and $\mathbf{CC1}[1]$.

References.

- [1] MCCALL, S., *Connexive implication*, *Journal of Symbolic Logic*, vol. 31 (1966), no. 3, pp. 415–433.
- [2] PRIEST, G., *Überconsistent logics and dialetheism*, *Crítica. Revista Hispanoamericana de Filosofía*, 2025.
- [3] WANSING, H., *Connexive modal logic*, *Advances in Modal Logic* (Renate Schmidt, Ian Pratt-Hartmann, Mark Reynolds and Heinrich Wansing, editors), College Publications, 2005, pp. 367–383.